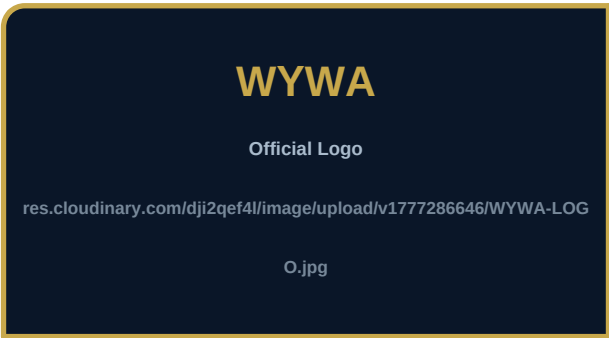




Logo URL: <https://res.cloudinary.com/dji2qef4l/image/upload/v1777286646/WYWA-LOGO.jpg>

WAZIRISTAN YOUTH WELFARE ASSOCIATION



Official Website & Fullstack System

Document Type	Project Requirements Specification (PRS)
Organization	Waziristan Youth Welfare Association (WYWA)
Location	South Waziristan, KP, Pakistan
Platform	VS Code + Blackbox AI (Fullstack Web Application)
AI Assistant	Blackbox AI (VS Code Extension)
Deployment	Vercel (Frontend) + Railway (Backend) + Neon (Database)
Version	v1.0 — April 2026

"Empowering Youth. Building Futures. Serving Waziristan."

1. EXECUTIVE SUMMARY

This document outlines the complete project requirements for the official website and fullstack digital platform of the Waziristan Youth Welfare Association (WYWA). The system will serve as WYWA's primary digital presence — a professional, fully dynamic, and content-managed platform that showcases the organization's programs, impact, team, and events, while enabling public engagement, volunteer management, and donation processing.

The platform will be built as a modern fullstack web application using a React/Next.js frontend and a Node.js/Express backend with a PostgreSQL database (Neon.tech). The system is designed for long-term scalability and easy content management by non-technical WYWA staff. Development is assisted using Blackbox AI in VS Code.

2. PROJECT OBJECTIVES

- Establish a professional, credible, and internationally presentable digital identity for WYWA.
 - Dynamically showcase all programs, events, news, and impact stories — manageable via an Admin CMS.
 - Enable online donation collection, volunteer applications, and membership registration.
 - Provide a secure Admin Dashboard for WYWA staff to manage all site content without coding.
 - Build a scalable, SEO-optimized, mobile-responsive platform aligned with NGO best practices.
 - Integrate social media feeds and multimedia (photos, videos) from WYWA's Facebook page and other channels.
 - Support bilingual content: English and Urdu/Pashto language toggle.
 - Provide transparency through a publicly accessible Annual Reports and Finance section.
-

3. RECOMMENDED TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React.js + Next.js 14	SSR, SEO, dynamic routing
Styling	Tailwind CSS	Responsive, utility-first design
Backend	Node.js + Express.js	REST API, business logic
Database	PostgreSQL (Neon.tech)	Persistent data storage
ORM	Prisma ORM	Type-safe DB queries
Auth	NextAuth.js / JWT	Admin login, role-based access
File Storage	Cloudinary	Image & video uploads
Email	Nodemailer + SendGrid	Contact forms, notifications
Payments	Stripe / JazzCash API / Bank Account	Online donations
Deployment	Vercel + Railway (Free Tier)	Hosting & CI/CD

CMS	Custom Admin Dashboard	Non-technical content management
Analytics	Google Analytics 4	Traffic & engagement tracking
AI Dev Tool	Blackbox AI (VS Code)	AI-assisted fullstack development

4. FUNCTIONAL REQUIREMENTS

4.1 Public Website — Pages & Features

- Homepage: Hero banner with animated stats, mission statement, featured programs, latest news, and testimonials
- About Us: History, vision, mission, values, organizational structure, legal registration details
- Programs: Dynamic listing of all active programs with categories, filters, and individual program detail pages
- Events: Calendar view + list view of upcoming and past events with registration forms
- Impact & Reports: Animated stats dashboard, beneficiary stories, annual reports (PDF downloads)
- News & Blog: Dynamic article system with categories, search, share buttons, and author info
- Gallery: Photo and video gallery organized by year and program — with lightbox viewer
- Team & Leadership: Dynamic team profiles with photos, roles, and bios
- Donate: Donation page with preset amounts, custom input, campaign selection, and payment gateway
- Volunteer: Application form with fields for skills, availability, and area of interest
- Contact Us: Contact form, embedded map, office details, social media links, and WhatsApp number
- Transparency: Financial statements, audit reports, registration certificates (public downloads)

4.2 Admin Dashboard (CMS)

- Secure login with role-based access (Super Admin, Editor, Viewer)
- Dashboard overview: visitor count, messages received, donations today, active programs
- Programs Manager: Add, edit, delete, publish/draft programs with rich text editor
- Events Manager: Create events, manage registrations, send email reminders
- News/Blog Manager: Full WYSIWYG editor, image upload, SEO meta fields, scheduling
- Team Manager: Add/edit/remove team members with photo upload
- Gallery Manager: Bulk upload photos, organize albums, add captions
- Donations Manager: View all donations, export CSV, filter by date/campaign
- Volunteer Applications: View, filter, approve/reject volunteer submissions
- Messages Inbox: View contact form submissions, mark as read/resolved
- Banner & Announcements Manager: Control hero slider content and urgent notices
- Reports Upload: Upload annual reports and finance documents (PDF)
- User Management: Add/remove admin users and assign roles
- SEO Settings: Per-page meta title, description, and Open Graph image
- Site Settings: Logo, contact info, social media links, footer content

4.3 User-Facing Features

- Newsletter signup with email confirmation
- Multilingual toggle (English / Urdu) with RTL support for Urdu
- Social media feed integration (Facebook page posts auto-displayed)
- Program search and filter by category, location, status
- Shareable content (programs, news, events) with Open Graph meta
- Donation receipt generation (PDF email) after successful payment
- Volunteer acknowledgment email upon form submission
- Printable / downloadable event certificates and reports
- Live chat / WhatsApp integration button for quick contact

5. DATABASE SCHEMA OVERVIEW

The following 13 core database models are required (PostgreSQL via Prisma ORM):

Model / Table	Key Fields
Users	id, name, email, password_hash, role, created_at
Programs	id, title, slug, description, category, status, image_url, start_date, created_at
Events	id, title, slug, description, date, location, image_url, is_published
EventRegistrations	id, event_id, name, email, phone, created_at
News	id, title, slug, body (rich text), author_id, category, image_url, published_at
TeamMembers	id, name, role, bio, image_url, order_index, is_active
Gallery	id, album_name, image_url, caption, program_id, uploaded_at
Donations	id, donor_name, email, amount, currency, campaign, payment_ref, created_at
Volunteers	id, name, email, phone, skills, availability, area, status, created_at
Messages	id, name, email, subject, message, is_read, created_at
Reports	id, title, year, file_url, category (financial/annual), created_at
SiteSettings	id, key, value (JSON blob for dynamic configuration)
Newsletter	id, email, is_confirmed, subscribed_at

6. CORE API ENDPOINTS

Module	Endpoints
Auth	POST /api/auth/login POST /api/auth/logout GET /api/auth/me

Programs	GET /api/programs GET /api/programs/:slug POST /api/programs (Admin) PUT /api/programs/:id DELETE /api/programs/:id
Events	GET /api/events GET /api/events/:slug POST /api/events/register POST /api/events (Admin)
News	GET /api/news GET /api/news/:slug POST /api/news (Admin)
Team	GET /api/team POST /api/team (Admin) PUT /api/team/:id (Admin)
Donations	POST /api/donations/initiate POST /api/donations/verify GET /api/donations (Admin)
Volunteers	POST /api/volunteers/apply GET /api/volunteers (Admin)
Messages	POST /api/contact GET /api/messages (Admin)
Gallery	GET /api/gallery POST /api/gallery/upload (Admin)
Reports	GET /api/reports POST /api/reports/upload (Admin)
Settings	GET /api/settings PUT /api/settings (Admin)

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Performance

- Page load time < 3 seconds on 4G mobile connection
- Google Lighthouse score ≥ 85 on Performance, Accessibility, and SEO
- Images served in WebP format with lazy loading
- API responses < 500ms for standard queries

7.2 Security

- All admin routes protected via JWT authentication middleware
- HTTPS enforced across all routes
- Input validation and SQL injection prevention (Prisma parameterized queries)
- CSRF protection on all mutation endpoints
- Rate limiting on public API endpoints (contact, donation, volunteer)
- Environment variables for all secrets — never hardcoded

7.3 Accessibility

- WCAG 2.1 Level AA compliance
- Keyboard navigable, screen reader compatible
- Sufficient color contrast ratios throughout
- Alt text required on all images via Admin CMS

7.4 SEO

- Dynamic meta tags per page (title, description, OG image)
- Structured data (Schema.org) for Organization, Events, and Articles
- Sitemap.xml auto-generated and submitted to Google Search Console

- Robots.txt configured appropriately
- Canonical URLs on all pages

7.5 Scalability & Reliability

- Stateless backend enabling horizontal scaling
- Database connection pooling
- Error monitoring via Sentry or similar
- Automated daily database backups
- 99.5% uptime target

8. PROJECT PHASES & ESTIMATED TIMELINE

Phase	Title	Timeline	Key Deliverables
Phase 1	Project Setup & Design	Week 1–2	Blackbox AI project init, tech stack setup, DB schema, wireframes, design system, color palette, component library
Phase 2	Core Public Pages	Week 3–5	Homepage, About, Programs listing & detail, Team, Contact — fully responsive with dummy data
Phase 3	Backend API & Database	Week 6–8	All REST API endpoints, database models, authentication, file upload (Cloudinary), email service
Phase 4	Admin Dashboard (CMS)	Week 9–11	Full CMS: login, programs/events/news/team/gallery managers, messages inbox, user roles
Phase 5	Advanced Features	Week 12–14	Donations & payment gateway, volunteer system, newsletter, multilingual (Urdu), social media integration
Phase 6	SEO, Performance & Security	Week 15	Lighthouse optimization, meta tags, sitemap, rate limiting, security audit, error monitoring
Phase 7	Testing & Launch	Week 16	End-to-end testing, UAT with WYWA team, bug fixes, domain setup, production deployment

9. RECOMMENDED FOLDER STRUCTURE (Blackbox AI / VS Code)

```

wywa-website/
├── frontend/ ← Next.js App
│   ├── app/
│   │   ├── (public)/ ← Public routes
│   │   │   ├── page.tsx ← Homepage
│   │   │   ├── about/
│   │   │   ├── programs/
│   │   │   ├── events/
│   │   │   ├── news/
│   │   │   └── gallery/

```

```
■ ■ ■ ■■ team/
■ ■ ■ ■■ donate/
■ ■ ■ ■■ volunteer/
■ ■ ■ ■■ contact/
■ ■ ■■ (admin)/ ← Admin Dashboard (protected)
■ ■ ■■ dashboard/
■ ■ ■■ programs/
■ ■ ■■ events/
■ ■ ■■ news/
■ ■ ■■ team/
■ ■ ■■ gallery/
■ ■ ■■ donations/
■ ■ ■■ volunteers/
■ ■ ■■ messages/
■ ■ ■■ settings/
■ ■■ components/
■ ■ ■■ ui/ ← Reusable UI components
■ ■ ■■ layout/ ← Navbar, Footer, Sidebar
■ ■ ■■ sections/ ← Page-specific sections
■ ■■ lib/ ← API calls, utils, hooks
■
■■■ backend/ ← Express.js API
■ ■■ routes/
■ ■■ controllers/
■ ■■ middleware/
■ ■■ models/ (Prisma)
■ ■■ utils/
■
■■■ prisma/
■ ■■ schema.prisma ← Database schema
■
■■■ public/ ← Static assets (logo, icons)
■■■ .env ← Environment variables
```

10. ENVIRONMENT VARIABLES REQUIRED

- DATABASE_URL — PostgreSQL connection string (from Neon.tech)
- NEXTAUTH_SECRET — NextAuth JWT secret key
- NEXTAUTH_URL — App base URL
- CLOUDINARY_CLOUD_NAME — Cloudinary account name
- CLOUDINARY_API_KEY + CLOUDINARY_API_SECRET — File upload credentials
- SENDGRID_API_KEY — Email service API key
- STRIPE_SECRET_KEY + STRIPE_PUBLISHABLE_KEY — Payment processing
- JAZZCASH_MERCHANT_ID + JAZZCASH_PASSWORD — Local PK payment option
- GOOGLE_ANALYTICS_ID — GA4 measurement ID

- SENTRY_DSN — Error monitoring
 - JWT_SECRET — Backend auth token signing
 - ADMIN_EMAIL — Default super admin email
-

11. DESIGN SYSTEM GUIDELINES

Color Palette

- Primary Navy: #0A1628 — backgrounds, headers, footer
- Primary Blue: #1A4A8A — buttons, accents, links
- Gold Accent: #C8A84B — highlights, calls-to-action, brand differentiator
- Off-White: #F8F9FC — section backgrounds
- Gray: #6B7A99 — body text, secondary content

Typography

- Display / Headings: Playfair Display (Google Fonts) — dignified, professional serif
- Body / UI: DM Sans — clean, modern, highly readable sans-serif
- Code / Mono: Courier New — for technical content
- Urdu/Pashto: Noto Nastaliq Urdu — RTL support

UI Principles

- Generous white space — professional NGO credibility
 - Consistent border-radius: 8–16px on cards and buttons
 - Subtle drop shadows on cards for depth
 - Smooth hover transitions (0.2–0.3s ease)
 - Scroll-triggered reveal animations (Intersection Observer)
 - Mobile-first responsive breakpoints: 375px, 768px, 1024px, 1280px
-

12. SUCCESS METRICS & ACCEPTANCE CRITERIA

- [AC-01] All 11 public pages are live, responsive, and SEO-tagged
- [AC-02] Admin CMS allows full CRUD on all content types without developer intervention
- [AC-03] At least one complete payment gateway (Stripe or JazzCash or Bank Account) functional end-to-end
- [AC-04] Contact form emails delivered to WYWA inbox reliably
- [AC-05] Volunteer and event registration forms capture and store data correctly
- [AC-06] Google Lighthouse score ≥ 85 on all key pages
- [AC-07] Urdu language toggle functional with correct RTL rendering
- [AC-08] Gallery uploads and displays images correctly via Cloudinary
- [AC-09] Admin roles enforced — editors cannot access user management or settings
- [AC-10] Platform passes security review: no exposed secrets, SQL injection prevention verified
- [AC-11] Full mobile responsiveness verified on iOS and Android browsers
- [AC-12] WYWA team can log in, create a new program, and publish it live within 5 minutes

Document prepared for WYWA Fullstack Website Project — Blackbox AI Development (VS Code). All specifications subject to revision based on stakeholder feedback.

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